

PNP Formal Reconstruction Report

Compiled Lean theorem inventory and current claim boundary

Coordinate PNP-FORMAL-RECONSTRUCTION-STATUS-2026-07-10-10

The repository does not currently establish
 $P = NP$.

Public theorem emission is disabled. The concrete publication gate is fail-closed because no audited concrete complexity target or reviewed kernel fingerprints exist in this reconstruction. The abstract string-handle bridge is not a standard P-versus-NP formalization and is never an activation source.

Compiled declarations	1761
Theorem-kind declarations	662
Assumption-free theorem-kind declarations	589
Project-specific axioms	5
Concrete publication gate passed	false

Inventory SHA-256: 4e4ab307d1651bb4440ab983595375a82cc172e418b8901901125d4b756f0b28

Generated from the pinned compiled Lean environment, not from source-text parsing.

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1 Current authority and claim boundary

This report is the current repository report surface. Its factual theorem inventory comes from the compiled PNP environment under `leanprover/lean4:v4.31.0`. The exporter enumerates `Environment.constants`, resolves each originating module, and calls `Lean.collectAxioms` for every exported project declaration. The committed inventory is `status/LEAN_THEOREM_INVENTORY.json`; the public mirror is byte-identical.

The target remains $P = NP$, but target wording is not theorem evidence. JavaScript acceptance, Boolean fields, JSON strings, historical release records, report prose, and external review cannot activate a theorem. Only the separately derived concrete publication gate may control theorem emission, and that gate is currently false.

Machine field	Current value
<code>mathematicalTheoremEstablished</code>	false
<code>publicTheoremEmissionAllowed</code>	false
<code>finalTheoremReady</code>	false
<code>publicTheoremStatement</code>	null
<code>rootLeanTheorem</code>	<code>PNP.Main.p_eq_np</code>
<code>rootLeanTheoremPresent</code>	false

2 Concrete publication gate

The compatibility entry is `PNP.Main.p_eq_np`; a matching name alone is insufficient. The separate publication target is `PNP.Main.ConcretePEqualsNP`. Eligibility also requires the exact theorem type, reviewed kernel fingerprints, and a tightly fixed Lean-standard axiom closure. No dummy concrete target is introduced by this report.

The expected fingerprints are intentionally unset. This is a fail-closed migration gate, not a missing runtime input. A verifier must reject any attempt to set `passed=true` while a fingerprint is null, while the concrete model is ineligible, or while any other subcheck is false.

Gate subcheck	Result
<code>standardComplexityModelEligible</code>	false
<code>concreteTargetPresent</code>	false
<code>concreteTargetIsDefinition</code>	false
<code>concreteTargetKernelTypeFingerprintConfigured</code>	false
<code>concreteTargetKernelTypeFingerprintMatches</code>	false
<code>concreteTargetKernelValueFingerprintConfigured</code>	false
<code>concreteTargetKernelValueFingerprintMatches</code>	false
<code>compatibilityRootPresent</code>	false
<code>compatibilityRootIsTheorem</code>	false
<code>compatibilityRootHasExactConcreteType</code>	false
<code>compatibilityRootKernelTypeFingerprintConfigured</code>	false
<code>compatibilityRootKernelTypeFingerprintMatches</code>	false
<code>axiomClosureFingerprintConfigured</code>	false
<code>axiomClosureFingerprintMatches</code>	false
<code>sourceClosureFingerprintConfigured</code>	false
<code>sourceClosureFingerprintMatches</code>	false
<code>axiomClosureUsesOnlyLeanStandardAllowlist</code>	false

Allowed foundation axioms are fixed to `Classical.choice`, `Quot.sound`, and `propext`. Project axioms, `sorryAx`, or any unknown axiom make the gate fail. The five currently disclosed project axioms are:

- `PNP.CheckPCCPackexp`
- `PNP.GeneratePCCPack`
- `PNP.LockedNANDThreshold`
- `PNP.ResidualBandExactMinimization`
- `PNP.SAT`

2.1 Why the abstract bridge is ineligible

The current `PNP.PEqualsNP` abbreviation compares witness classes whose language and algorithm structures carry string names or code handles rather than concrete execution semantics and proved polynomial bounds. Consequently, even a kernel-clean theorem of that abstract type would not establish the standard P-versus-NP statement. It is compatibility information only.

3 Formal milestone ledger

Each earned row requires exact compiled names and theorem kinds, empty axiom closures, per-name domain-separated kernel-type SHA-256 matches for 22 detailed candidates, and the pinned whole-Lean source closure. Human-readable scope remains conservative: type or source drift revokes credit.

Milestone	Status	Exact scope	Boundary
Typed direct-wire NAND semantics	formalized	Typed topological Boolean NAND programs and ordered multi-output direct-wire semantics.	This does not establish circuit minimization, SAT, or $P = NP$.
Finite enumeration, equivalence, and reference minimum	formalized	Exhaustive finite Boolean direct-wire search under the empty-profile reference model.	No polynomial-runtime result is obtained from this exhaustive reference search.
Concrete framed replacement and slack	formalized	The concrete serial framed context with support outputs and explicit bypass wires.	This is not an arbitrary-support replacement theorem for the report's global family.
Locked-NAND local candidates and baseline accounting	formalized	Typed local macro candidates, source-derived counts, and five finite local square baselines.	Local minima are not a global BaselineDistinct or locked-NAND threshold theorem.
Conditional locked-NAND threshold boundary	formalized-with-premises	A proof-bearing six-premise candidate boundary for an arbitrary satisfiable proposition.	The premises are not instantiated by a uniform SAT-to-locked-NAND builder.
Fail-closed explicit-list residual routes	formalized-explicit-list-only	Executable strict-gain search over a caller-supplied finite implementation list.	Unresolved excludes no gain outside the supplied list and cannot imply ZeroSlack.
Global locked-NAND construction and threshold	not-formalized	A uniform answer-independent SAT instance builder, global baseline, trace equivalence, and threshold.	The conditional boundary does not discharge this missing theorem.

Milestone	Status	Exact scope	Boundary
Global ZeroSlack, PCCMin, and polynomial runtime	not-formalized	Complete residual routing, global ZeroSlack contradiction, exact minimization, and polynomial bounds.	Proof-bearing result structures and explicit-list failure do not supply global completeness.
Concrete standard P-versus-NP target and root theorem	not-formalized	Concrete machine semantics, complexity classes, SAT, reductions, and the publication root theorem.	The current string-handle PNP.PEqualsNP bridge is nonstandard and publication-ineligible.

4 Compiled inventory summary

The inventory contains 1761 exported `PNP.*` kernel declarations from 22 modules. It excludes 33 private compiler auxiliaries. Counts describe the compiled environment; they do not measure mathematical completeness.

Module	Declarations	Theorems
PNP.Bridge	93	22
PNP.Complexity	93	20
PNP.DirectWireBaseline	48	25
PNP.LockedNAND	11	4
PNP.LockedNANDBaseline	57	22
PNP.LockedNANDDirect	84	57
PNP.LockedNANDLocalBaseline	30	22
PNP.LockedNANDMacros	288	98
PNP.LockedNANDPrefix	84	40
PNP.LockedNANDThresholdBoundary	60	35
PNP.Main	36	12
PNP.NANDComposition	77	38
PNP.NANDEnumerator	120	46
PNP.NANDMinimum	56	23
PNP.NANDSemantics	198	80
PNP.NANDSlack	15	12
PNP.NANDTruthTable	72	28
PNP.PCCMin	44	8
PNP.ResidualBand	9	2
PNP.ResidualRoutes	141	46
PNP.SAT	4	1
PNP.ZeroSlack	141	21

5 Remaining formal blockers

Seven formal blockers remain active:

1. `Formal.ConcreteComplexityModel`
2. `Formal.ConcreteSAT`
3. `Formal.LockedNANDThreshold`
4. `Formal.ResidualBandMinimizer`
5. `Formal.ZeroSlack`
6. `Formal.PolynomialRuntimeAndCertificateBounds`
7. `Formal.RootTheoremAndAxiomAudit`

The conditional locked-NAND threshold module assumes typed baseline and full candidates, baseline conditions, preservation of existing outputs, an unsatisfiable final-zero law, and satisfiable final-output

laws. The explicit residual scanner searches only a caller-supplied finite list. Neither result constructs the missing global reduction or proves global ZeroSlack, polynomial PCCMin, SAT in P, or P equals NP.

6 Historical report provenance

Earlier 56-page report revisions stated a direct checker-mediated P-versus-NP claim. Those bytes remain historical audit material at their pinned legacy coordinates and through the explicit archive and supersession records. They are not imported into, quoted as authority by, or used to generate this report. File identity and replay of historical predicates do not establish the named mathematical propositions.

Source tag	final-pnp-proof-report-hardened-7072f8d
Source commit	7072f8d0bda6d44d240f9bb3fad624fd357e1278
Archive manifest	archive/legacy-v0/ARCHIVE.json

7 Verification

The current reconstruction can be checked with:

```
lake build PNP
node scripts/export-lean-theorem-inventory.mjs --check
node scripts/generate-formal-publication.mjs --check
node pcc-formal-reconstruction-status0.mjs --json
npm run pnp:verify -- --no-write
npm run report:check
```

The inventory coordinate is PNP-LEAN-THEOREM-INVENTORY-2026-07-10-10. Its exact byte digest is 4e4ab307d1651bb4440ab983595375a82cc172e418b8901901125d4b756f0b28. The report is regenerated from that inventory and the reviewed publication map. The reviewed milestone source-closure SHA-256 is 0c96305d4f2235aae359c865e91e0f58cafa4d20d9b040a16e992846360cfa3f; the computed and pinned values match. Successful regeneration confirms consistency of these status surfaces; it does not turn an absent concrete theorem into a proof.